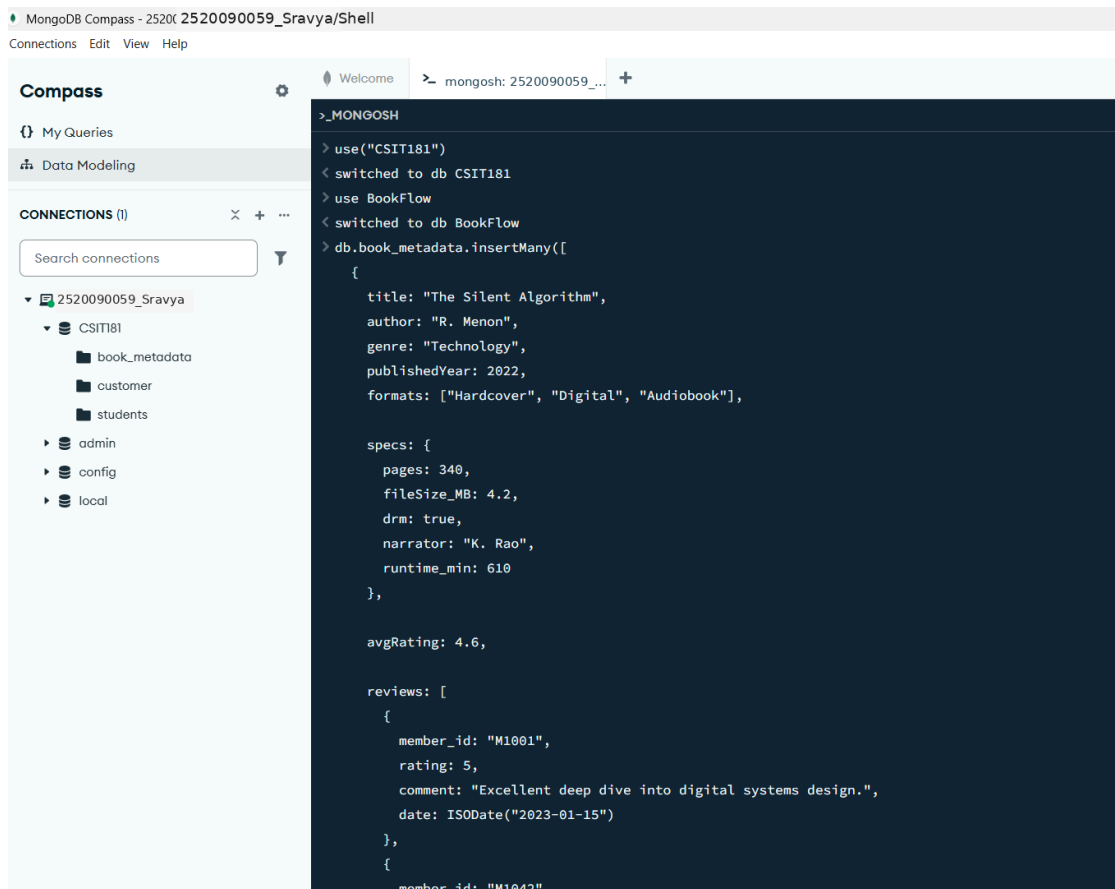


Week -3

Task-1



```
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6a8d21529d8318f65ed50b11'),
      '1': ObjectId('6a8d21529d8318f65ed50b12'),
      '2': ObjectId('6a8d21529d8318f65ed50b13'),
      '3': ObjectId('6a8d21529d8318f65ed50b14')
    }
  }
}
BookFlow>
```

```

> db.book_metadata.find().pretty()
< {
  _id: ObjectId('6a8d21529d8318f65ed50b11'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',

```

```

      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b12'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',

```

```

        date: 2022-06-01T00:00:00.000Z
      }
    ]
  }
  {
    _id: ObjectId('6a8d21529d8318f65ed50b13'),
    title: 'Whispers of Yesterday',
    author: 'S. Kapoor',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [
      'Hardcover',
      'Paperback'
    ],
    specs: {
      pages: 410,
      binding: 'Sewn'
    },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: 2020-01-15T00:00:00.000Z
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.',
        date: 2020-02-11T00:00:00.000Z
      }
    ]
  }
]

```

```

    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [
      'Digital',
      'Audiobook'
    ],
    specs: {
      fileSize_MB: 5.6,
      drm: true,
      narrator: 'P. Iyer',
      runtime_min: 540
    },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: 2023-08-01T00:00:00.000Z
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: 2023-09-14T00:00:00.000Z
      }
    ]
  }
]
}

```

Task-2

a) Books rated above 4 (\$gt)

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
BookFlow>
```

b) Books available in specific formats (\$in)

```
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
BookFlow> |
```

c) Combine both with \$and (implicit)

```
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d21529d8318f65ed50b11'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
  ],
}
```

```
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.',
    date: 2023-02-20T00:00:00.000Z
  }
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b12'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    },
  ],
}
```

```
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: 2022-06-01T00:00:00.000Z
  }
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b14'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: 2023-08-01T00:00:00.000Z
    },
  ],
}
```

```
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: 2023-09-14T00:00:00.000Z
    }
  ],
}
}
BookFlow >
```

d) Regex query — books with "Digital" format

```
> db.book_metadata.find(
  {
    formats: {
      $regex: "Digital",
      $options: "i"
    }
  },
  {
    title: 1,
    formats: 1
  }
)
< {
  _id: ObjectId('6a8d21529d8318f65ed50b11'),
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b12'),
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b14'),
```

```
  _id: ObjectId('6a8d21529d8318f65ed50b14'),
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
BookFlow> |
```

e) Regex query — reviews mentioning "Digital" in comments (nested field)

```
> db.book_metadata.find(
  {
    "reviews.comment": {
      $regex: "Digital",
      $options: "i"
    }
  },
  {
    title: 1,
    "reviews.$": 1
  }
)
< {
  _id: ObjectId('6a8d21529d8318f65ed50b11'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b12'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
```

```
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b14'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: 2023-08-01T00:00:00.000Z
    }
  ]
}
BookFlow> |
```

Task-3

```
> {
  $match: {
    publishedYear: { $gt: 2020 }
  }
}
< 2020
BookFlow> |
```

```
> {
  $unwind: "$reviews"
}
< $reviews
BookFlow>
```

```
> db.book_metadata.aggregate([
  {
    $match: {
      publishedYear: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      avgReviewRating: {
        $avg: "$reviews.rating"
      },
      totalReviews: {
        $sum: 1
      },
      author: {
        $first: "$author"
      }
    }
  },
  {
    $sort: {
      avgReviewRating: -1
    }
  }
])
```

```
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: {
      $round: ["$avgReviewRating", 2]
    },
    totalReviews: 1
  }
}
])
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
BookFlow> |
```

TASK-4

```
> db.book_metadata.find({
  $text: {
    $search: "Digital"
  }
})
< {
  _id: ObjectId('6a8d21529d8318f65ed50b11'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
  ],
}
```

```
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ],
  _id: ObjectId('6a8d21529d8318f65ed50b14'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: 2023-08-01T00:00:00.000Z
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: 2023-09-14T00:00:00.000Z
    }
  ],
  _id: ObjectId('6a8d21529d8318f65ed50b12'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: 2022-06-01T00:00:00.000Z
    }
  ],
}
```

```
avgRating: 4.9,
reviews: [
  {
    member_id: 'M4001',
    rating: 5,
    comment: 'The Digital format made annotations so easy.',
    date: 2023-08-01T00:00:00.000Z
  },
  {
    member_id: 'M4002',
    rating: 5,
    comment: 'Mind-bending and well narrated.',
    date: 2023-09-14T00:00:00.000Z
  }
]
}
{
  _id: ObjectId('6a8d21529d8318f65ed50b12'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: 2022-06-01T00:00:00.000Z
    }
  ],
}
```

```
avgRating: 4.2,
reviews: [
  {
    member_id: 'M2001',
    rating: 4,
    comment: 'Solid Digital read, clean layout.',
    date: 2022-05-10T00:00:00.000Z
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: 2022-06-01T00:00:00.000Z
  }
]
}
BookFlow >
```



```

> db.book_metadata.find({
  $text: {
    $search: "Digital"
  }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
      indexPrefix: {}
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 7,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [
          'digit'
        ],
        negatedTerms: [],
        phrases: []
      },
      negatedTerms: [],
      phrases: []
    }
  }
}

```

```

  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    nss: 'BookFlow.book_metadata',
    inputStage: {
      stage: 'IXSCAN',
      nss: 'BookFlow.book_metadata',
      keyPattern: {
        _fts: 'text',
        _ftsx: 1
      },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {}
    }
  }
}

```

```

    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 7,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [
          'digit'
        ],
        negatedTerms: [],
        phrases: []
      },
      negatedTerms: [],
      phrases: []
    }
  }
}

```

```

  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  docsRejected: 0,
  inputStage: {
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0
    }
  }
}

```

```

    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 3,
    seeks: 1,
    dupsTested: 3,
    dupsDropped: 0,
    peakTrackedMemBytes: 26
  }
}

```

```

queryShapeHash: '69E5F591B2A8041B301CE1729E6C89F575F3AC5A7AB1AFB6977B0349FD8C89E5',
peakTrackedMemBytes: Long('26'),
command: {
  find: 'book_metadata',
  filter: {
    '$text': {
      '$search': 'Digital'
    }
  },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'LAPTOP-41COT5AA',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```

```

> db.book_metadata.find({
  "reviews.comment": {
    $regex: "Digital",
    $options: "i"
  }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
      nss: 'BookFlow.book_metadata',
      direction: 'forward'
    }
  }
}

```

```

    nss: 'BookFlow.book_metadata',
    direction: 'forward'
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 0,
  totalKeysExamined: 0,
  totalDocsExamined: 4,
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    }
  },
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 5,
  advanced: 3,
  needTime: 1,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'BookFlow.book_metadata',
  direction: 'forward',
  docsExamined: 4
}
}

```

```

queryShapeHash: '5325C922132166A4CE16461976731802F7654821C8246E8E6E2F7BD579A AFC86',
command: {
  find: 'book_metadata',
  filter: {
    'reviews.comment': BSONRegExp('Digital', 'i')
  },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'LAPTOP-41COT5AA',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
BookFlow>

```

```

> db.book_metadata.createIndex({ formats: 1 })

db.book_metadata.createIndex({ avgRating: -1 })

db.book_metadata.find({
  avgRating: { $gt: 4 }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'BookFlow.book_metadata',
      inputStage: {

```

```

stage: 'IXSCAN',
nss: 'BookFlow.book_metadata',
keyPattern: {
  avgRating: -1
},
indexName: 'avgRating_-1',
isMultiKey: false,
multiKeyPaths: {
  avgRating: []
},
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'forward',
indexBounds: {
  avgRating: [
    '[inf, 4)'
  ]
}
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 3,
  totalKeysExamined: 3,
  totalDocsExamined: 3,

```

```

queryShapeHash: '79E1597FBF1E7E0551896CC4A5A0604E417E5E37F937A039A8C2D303FB513A0',
command: {
  find: 'book_metadata',
  filter: {
    avgRating: {
      '$gt': 4
    }
  },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'LAPTOP-41COT5AA',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
BookFlow>

```